



Important information

- Hypothermia is core temperature of body $>40.5^{\circ}\text{C}$
- Altered mentation + core temperature $>40.5^{\circ}\text{C}$: **Heat stroke**

MILD HYPOTHERMIA	32-35°C
MODERATE HYPOTHERMIA	28-32°C
SEVERE HYPOTHERMIA	$<28^{\circ}\text{C}$
POIKILOTHERMIA • Altered mentation: HACE+ Concomitant hypothermia	$<32^{\circ}\text{C}$

FEATURES SEEN IN HYPOTHERMIA

- Paradoxical undressing (taking off clothes, burying in ice) due to altered mentation
- ECG:
 - Presence of **J waves**
 - T wave inversion due to myocardial hypoxia $\rightarrow$ coronary blood flow $\downarrow$
- Core temperature probe placement sites
 - **Lower esophagus**: Preferred site
 $\rightarrow$ Incisor teeth to Lower esophageal sphincter: **$<40\text{ cm}$**
 - Ideal site: Pulmonary artery with Swan Ganz catheter
 - Rectum: 15 cm depth and must touch the rectal wall



ON ECG

- If **ventricular fibrillation** is detected: Immediate DC shock, but next DC shock should be deferred and first do **rewarming** and $\uparrow$ temperature by $1-2^{\circ}\text{C}$

REWARMING APPROACHES

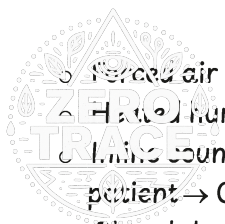
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PASSIVE EXTERNAL REWARMING

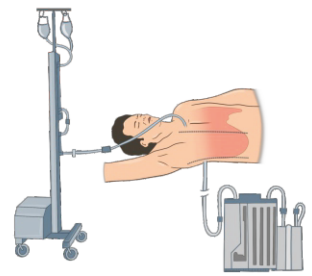
- Indicated in previously healthy person with accidental hypothermia: **Cover and insulate** patient in warm environment
- Rate of rewarming: **$0.5^{\circ}\text{C}-2^{\circ}\text{C}/\text{hour}$**
 - Heat pads should not be put directly in contact with the skin

ACTIVE REWARMING

- Indications:
 - **Core temperature $<32^{\circ}\text{C}$**
 - CV instability (HR $\downarrow$, CO $\downarrow$, BP $\downarrow$)
 - CNS malfunction
- Methods:



- Forced air heating blanket/circulating hot water
- Humidified oxygen: **Mask/Endotracheal tube**
- Lining countercurrent heat exchanger: **Warm saline** to patient → Cardiac contractility starts improving
- Closed thoracic lavage: If chest wall is frozen
- Peritoneal lavage with warm saline
- Extracorporeal rewarming:



00:38:14

EXTRACORPOREAL REWARMING

- Rate of rewarming (ROR)
 - Continuous venovenous rewarming: 2-3°C
 - Haemodialysis: 3-4°C
 - Cardiopulmonary bypass: 9.5°C

Important Information

- Frost Bite – Frozen tissue immersion in circulating water at **37°C - 40°C**



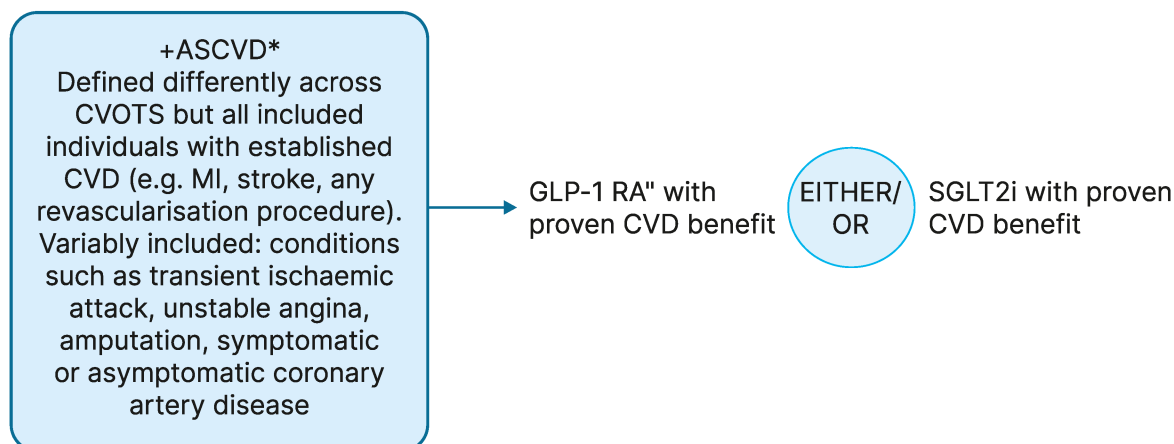
33. DIABETES MELLITUS PART - 1



BASICS OF DIABETES MELLITUS

00:01:53

- Leading cause of death - ↑ Cardiovascular mortality
 - DM contributes to accelerated atherosclerosis.
- Sudden cause of death in Diabetes mellitus - Silent MI
- Drugs to lower CV mortality in Diabetes mellitus
 - GLP-1 receptor agonist like **Semaglutide**
 - SGLT2i like **Empagliflozin/ Canagliflozin** → Cause salt loss from body



- **M/c type of Diabetes** - Type 2 DM d/t insulin resistance
 - Cause - Polygenetic inheritance + Multifactorial

INDICES TO MEASURE INSULIN RESISTANCE FOR CLINICAL PURPOSES & PREDICT T2DM

00:09:39

- **HOMA-IR: Homeostasis Model for Assessment of Insulin Resistance:** $\text{Fasting glucose (mmol/L)} \times \text{Fasting insulin (mIU/L)} / 22.5$
 - HOMA-IR - > 2 → indicates insulin resistance
 - Fasting insulin - measured by RIA (Radioimmunoassay)
- **QUICKI** (Quantitative insulin sensitivity check index)
- **Mitsuda index**

ACANTHOSIS NIGRICANS

00:12:16

- Hyperpigmentation over the dorsum of neck/axilla
 - Causes
 - Type 2 DM
 - PCOD
 - Metabolic syndrome
 - Ca Pancreas & Ca colon
 - Stomach cancer
- } Sign of Internal malignancy



Dorsum neck/axilla

CLASSIC SYMPTOMS

00:14:06

1. Polyuria - Urine output >3L/day
 2. Polydipsia - Water intake > 6L/day
 3. Weight loss
 4. Polyphagia - Not a classic symptom (not measurable)
 5. UTI/Vulvovaginitis - Recurrent
 6. Asymptomatic
- } Classic symptoms (measurable)

ADA 2023: DIAGNOSTIC CRITERIA OF DIABETES MELLITUS

00:17:33

	NORMAL (mg/dL)	PRE-DIABETIC (mg/dL)	DM (mg/dL)
Fasting plasma glucose (mg/dL)	<100	100-125 (Impaired fasting glucose)	≥126
2-hr plasma glucose (After the patient consumes 75 grams of glucose)	<140	140-199 (Impaired glucose tolerance)	>200

- Post prandial value: Useful for monitoring of DM not for diagnosis of DM
- Random blood sugar: Can be useful for diagnosis in presence of symptoms

TYPES OF DIABETES MELLITUS

00:26:28

TYPE 1 DIABETES MELLITUS	• Etiology - Autoimmune destruction of β cells ◦ Anti islet cell antibody ◦ 80% destruction → Insulinopenia • ↓ Radioimmunoassay insulin levels • Initially asymptomatic, subsequently become symptomatic • Fasting plasma glucose ↑ • Stage 3 Type 1 Diabetes mellitus → Autoimmune dysglycemia
TYPE 1.5 DIABETES MELLITUS	• LADA: Latent autoimmune diabetes in adults • Type of DM1 from etiology purposes • Slow in presentation
TYPE 2 DIABETES MELLITUS	• Polygenetic inheritance • Insulin resistant • Elevated insulin levels

SPECIFIC TYPES OF DIABETES MELLITUS

00:30:35

Monogenic	• Mature Onset Diabetes in Young (MODY) • Autosomal dominant inheritance
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Damage to exocrine component Malabsorptions/ Exocrinopathies	• Cystic fibrosis • Chronic pancreatitis	} Malabsorption
Endocrinopathies	• Acromegaly • Cushing syndrome • Pheochromocytoma • Glucagonoma	
Drugs	• Steroids • mTOR inhibitors • Calcineurin inhibitors • Thiazides • Propranolol • Antipsychotics • Statins	

ADA: DIAGNOSIS OF DIABETES MELLITUS

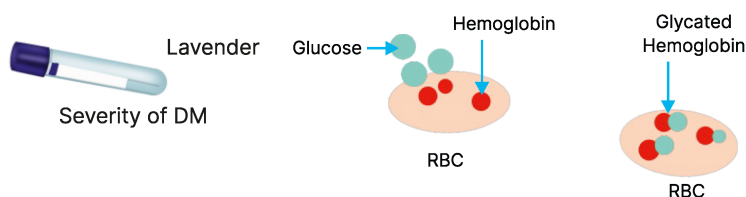
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- Symptoms of hyperglycemia/hyperglycemic crisis + RBS > 200 mg/dL
- Asymptomatic patient + Fasting plasma glucose > 126 mg/dL (> 7 mmol/L) & 2-hour plasma glucose > 200 mg/dL (>11.1 mmol/L).
- Asymptomatic patient + HbA1c ≥ 6.5%
- Post prandial value – Used for Monitoring/Follow up of the patient

HbA1C

00:42:24

- Lavender top - Used for HbA1c; Normal ≤ 5.6%
- Determines the severity of DM
- Gives long-term control of Diabetes
- Gives Estimated average blood glucose value for 8-12 weeks
 - $28.7 \times \text{HbA1C} - 46.7$ or
 - HbA1C value $\times 21$ (not used regularly)
- Retrospective test: Not affected by recent food intake.
- **IOC/Best for Diabetes**
- Not useful for the diagnosis of bronze diabetes.



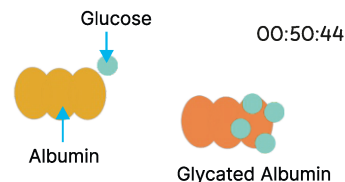
BRONZE DIABETES

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- When PRBC transfusion is given to Thalassemia patients (↓ HbA) → Iron toxicity
 - Damage to Liver → **Cirrhosis**
 - Damage to Pancreas → **Insulinopenia**
 - **Bronze skin** (Iron deposition in skin)

SEVEN FRUCTOSAMINE/GLYCATED ALBUMIN

- At least for **short-term control** of DM
- Estimate average glucose of previous 2-3 weeks
- Based on glycation of proteins



HIGH RISK POPULATION FOR SCREENING OF DM

- **Obesity (BMI > 30)/Overweight (BMI: > 25)**
- Physical inactivity
- Family history of DM
- History of CVD
- HTN or on medication: >130/80mmHg
- HDL <35 or TG's >250mg/dL
- PCOS/Acanthosis nigricans

00:52:04

SCREENING FREQUENCY IN VARIOUS GROUPS

00:53:46

CRITERIA	SCREENING FREQUENCY
Impaired glucose tolerance, impaired fasting glucose (100-125), HbA1C: 5.7-6.4% (Pre-diabetic)	• Yearly basis
Gestational DM	• 3 yearly basis
General population	• Screening is started ≥ 35 years - According to ADA • ≥ 45 years: According to Harrison 21st edition

REMISSION OF DIABETES MELLITUS

00:55:37

- Definition - HbA1C <6.5% $\times$ 3 months without the use of oral hypoglycemic agents/Insulin

PATIENT CHARACTERISTICS

00:57:55

- 25-59 years with BMI ≥ 25
 - Fasting plasma glucose >110 mg
 - HbA1c > 6%
 - GDM history
 - Remission can be achieved by
 - Metformin
 - Hypocaloric diet with Weight loss > 7%
 - Pioglitazone
- } $\uparrow$ chance of Diabetes mellitus

LONG-TERM COMPLICATIONS OF DIABETES MELLITUS

- Continuous glucose monitoring devices: Hope to decrease long term complication of DM



MICROVASCULAR VS MACROVASCULAR COMPLICATIONS OF DIABETES MELLITUS

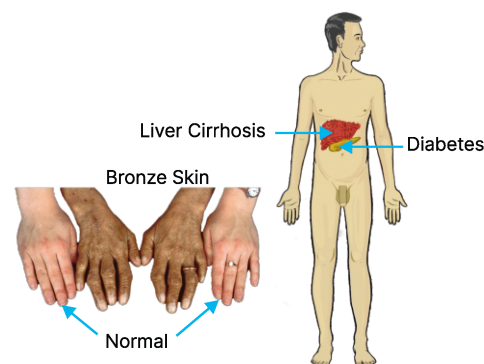
MICROVASCULAR COMPLICATIONS	MACROVASCULAR COMPLICATIONS
M/c complication of DM (overall) - Diabetic neuropathy → damage to small fibres (small unmyelinated C fibres)	Coronary artery disease → Accelerated atherosclerosis
- Diabetic retinopathy & Diabetic nephropathy (ESRD/Allogenic kidney transplant) - Progression - T1 DM → 5 years to develop - T2 DM → 15 to 20 years to develop - Diabetic nephropathy → ESRD/Allogenic kidney transplant - Chance of blindness is 25 times higher than the normal population.	- Peripheral arterial disease - ↑ risk of digital gangrene - Incidence of PAD in DM is 100 times higher as compared to the normal population
Best intervention → Strict HbA1c control	Best intervention → Strict BP control

ETIOLOGY OF TYPES OF DIABETES MELLITUS

01:07:31

T1 DM

- Pediatric onset
- Autoimmunity:** By HLADQ1 0301 and 0302 (M/c)
- ↓ RIA insulin levels
- Fulminant diabetes: Viral insult to pancreas - Mumps/Rubella
- Bronze diabetes (Child with thalassemia given packed RBCs without iron chelators)
 - Bronzing skin
 - Beta cell damage → DM1
 - Cirrhosis of liver
- Autoimmune polyendocrine syndrome (Type 1 & 2)
- Defective CTL-4A gene



STAGES OF TYPE 1 DIABETES MELLITUS

Stage 1 - Asymptomatic, Normoglycemia

Stage 2 - Asymptomatic, Dysglycemia

Stage 3 - Symptomatic, Hyperglycemia

	STAGE 1	STAGE 2	STAGE 3
B-cell autoimmunity (with the presence of autoantibodies)	Present	Present	Present



β-cell loss	Present	Present	Present
Dysglycemia	Absent	Hyperglycemia	Hyperglycemia
Symptoms	Absent	Absent	Present

TYPE 2 DM

- Polygenetic inheritance } ↑ levels of adipokines
- Centripetal Obesity } **Adiponectin** ↓
- **Adiponectin**: Guardian angel against obesity

TYPE 1.5 DM

- LADA: Latent autoimmune diabetes in adults
 - Etiology - Autoimmunity
 - Age-wise presentation - Adult onset
 - Antibody responsible: Anti islet cell antibody / **Anti GAD** (Glutamine acid decarboxylase)
 - **C-Peptide levels**
 - <0.3 mmol/l - T/t with Insulin
 - >0.7 mmol/L - T/t with Metformin
- Sulfonylureas are not recommended**

MODY (MATURITY ONSET DIABETES OF YOUNG)

01:17:37

- Rare

CLINICAL FEATURES SUGGESTIVE FOR MODY

ANY OF THE FOLLOWING

- Diabetes diagnosed ≤ 30 years
- Diabetes diagnosed ≤ 45 years in people without obesity/insulin resistance/metabolic syndrome
- Diabetes diagnosed ≤ 45 years and a family history of diabetes in ≥ **2 generations** in an autosomal dominant fashion

IN THE ABSENCE OF

- Diabetic ketoacidosis
- Pancreatic islet autoantibodies
- No endogenous insulin production outside the honeymoon period of about 3 years (e.g., undetectable C peptide)

Inheritance	• Autosomal dominant
Peak age of onset	• 25 years
Presentation	• Asymptomatic + ≥2 generations with history of DM
Types	• 14



M/c subset

- MODY 3 (India/USA)
- Gene responsible: HNF-1 α gene on chromosome 12

Diagnosis

- Genetic testing

First line intervention

- **Sulfonylureas** as β cell mass is normal

SUMMARY

01:22:28

	T1DM	T2DM	LADA (β cell mass $\downarrow$)	MODY
Presentation	Childhood	Adult	Adult	Adult
Etiology	Autoimmune	Polygenetic	Autoimmune	AD
Family H/O	<15%	>50%	Rare	>90%
Islet cell antibodies	+	-	+	-
1st line management	Insulin	Metformin + GLP-1 RA	Depending on C-peptide levels • <0.3 mmol/L: Insulin • >0.7 mmol/L: Metformin	Glipizide



Q) Which of the following tests is best for assessing long-term control of Diabetes mellitus?

- a. Glycosylated hemoglobin
- b. Serum Fructosamine
- c. Glycated albumin
- d. Post prandial blood glucose

Ans. (a)

Q) In a patient with T1DM at the 3rd stage of beta cell damage, which of the following conditions is most likely to be observed?

- a. Autoimmune positive + Normoglycemia
- b. Autoimmune negative + Normoglycemia
- c. Autoimmune positive + Dysglycemia
- d. Autoimmune negative + Dysglycemia

Ans. (c)



37. DIABETES MELLITUS PART-2



CASE SCENARIO

00:00:54

An 8-year-old child was admitted with a diagnosis of Pneumonia and is on IV Antibiotics for 3 days. Since last night, he has been having multiple vomiting episodes and abdominal pain, and his mother says he is very drowsy and does not recognise his mother. Vitals: Pulse - 110/min, BP - 80/60 mmHg, RR- 20/min, and epigastric tenderness is noted.

- SpO₂ on room air - 98%
- ABG was done - pH ↓
- pCO₂ - ↓
- HCO₃ - ↓
- Random blood sugar - 300 mg/dL
- Urine for ketone: 4+
- Plasma beta-hydroxybutyrate: ↑
- Clinical diagnosis: **Diabetic Ketoacidosis**
- **M/c cause of death in DKA (in child) - Cerebral edema**

Pointers that may not be given in question	Kussmaul breathing, Fruity odor in breath
IOC for DKA	Plasma β-OH butyrate levels - ↑
First line for DKA	NS+KCl (40 mEq/L)
TOC for DKA	Insulin drip
Fluid used when blood sugar falls to < 250 mg/dL	5% Dextrose + 0.45% NS
Cause of death	Cerebral oedema > Dehydration
Trigger	Pneumonia, UTI, T1DM: Insulin pump malfunction: Insulin (non-compliance)

LONG-TERM MANAGEMENT OF T1DM

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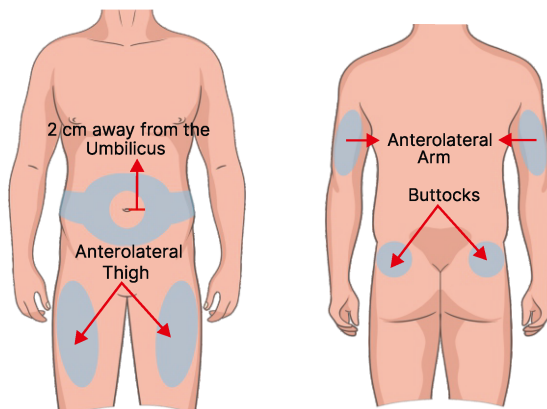
1. CARBOHYDRATE COUNTING	• T2DM: Caloric restriction
2. MODERATE TO VIGOROUS PHYSICAL ACTIVITY	• 60 min/ day



- DNA Recombinant Technology: Human Insulin
- Preferred route - **Subcutaneous**
- Preferred site - Anterior abdominal wall
- Preferred method of delivery - Continuous Subcutaneous Insulin Infusion (Insulin pump)

SITE FOR INSULIN DELIVERY

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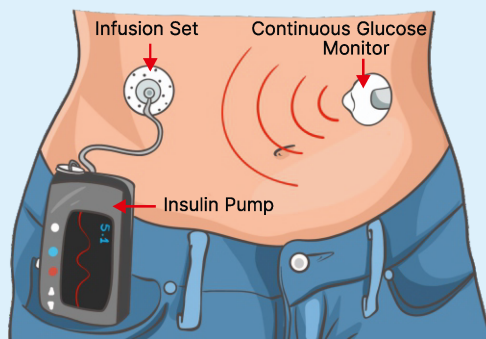


- Preferred site: Anterior abdominal wall (2 cm away from the umbilicus)
 - Anterolateral thigh
 - Anterolateral arm
 - Buttocks
- Rotate the sites of Insulin administration to prevent lipodystrophy

INSULIN DELIVERY METHODS

00:22:31

1. CSSI (CONTINUOUS SUBCUTANEOUS INSULIN INFUSION)



- Preferred site of delivery
- Continuously rotate the Insulin to reduce Lipodystrophy
- Very expensive

2. INSULIN PEN



- 32G needle
- Injection is done at 90 degrees to skin
- Length is 4.5 mm - 12.7 mm
- Almost painless delivery, easy to carry



3. INHALED INSULIN



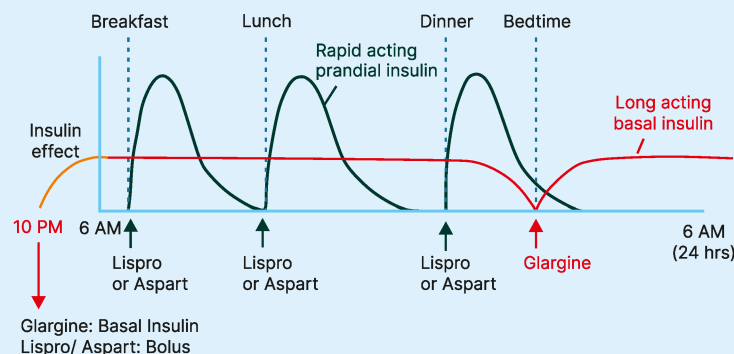
- Rapid acting
- Onset of action is < 30 minutes

MULTI-DOSE VIAL OF INSULIN



- Storage temperature: + 2-8 degrees Celsius
- Shelf life - Use within 4 weeks
- Self-injection

BASAL BOLUS REGIMEN



- Basal: Long-acting basal insulin (Glargine) - 50% total calculated dose
- Lispro/Aspart (Bolus insulin) is given 25 minutes before the meal → Short-acting Insulin: 50% total calculated dose
- Peakless insulin: Glargine

SIDE EFFECTS OF INSULIN USE

00:34:34

- Lipodystrophy affects insulin absorption and affects glycemic control
- Hypoglycemic: Plasma glucose < 54 mg/dL

Level 1	55-70 mg/dL
Level 2	< 54 mg/dL
Level 3	Any level that results in the incapacitation of the patient

SYMPTOMS OF HYPOGLYCEMIA

00:36:28

- Mood swings
- Attention span ↓
- Palpitations
- Tremors



TREATMENT OF SYMPTOMATIC HYPOGLYCEMIA

00:39:10

- **IV: 25-50% Dextrose** (Child- 10% Dextrose)
- **Inj glucagon S/C** (Failure to get IV access)
- These are given to prevent seizures

ROLE OF BETA BLOCKERS IN DM

00:41:01

- Mask the symptoms of hypoglycemia except Diaphoresis
- DM + ASCVD, Chronic stable angina → **Metoprolol**

SOMOGYI PHENOMENON

00:44:36

10 pm	4 am: Blood sugar begins to fall to low levels	7 am: 200 mg/dL
• Basal dose of Insulin is given (10 units Glargine) • Blood sugar level - 150 mg/dL • Due to the administration of the extra dose of insulin, blood sugar falls	• Nocturnal hypoglycemia • Causes counter-regulatory hormones like Glucagon, cortisol, and catecholamines to begin to rise • Blood sugar level - 45 mg/dL	• Early morning hyperglycemia - 200mg/dL

DAWN PHENOMENON

00:49:37

10 pm	4 am: Blood sugar begins to fall to low levels	7 am: 200 mg/dL
• Blood sugar level - 150 mg/dL • Basal dose of Insulin is given (10 units Glargine)	• GLUT4 receptors exhibit downregulation between 4 am and 7 am • Blood sugar level - 150 mg/dL • Nocturnal hyperglycemia	• Early morning hyperglycemia - 200mg/dL

PROPERTIES OF INSULIN PREPARATION

00:52:39

PROPERTIES OF INSULIN PREPARATIONS			
PREPARATION	TIME OF ACTION		
	ONSET, h	PEAK, h	EFFECTIVE DURATION, h
LONG-ACTING			
Degludec (Longest)	1-9	Peakless Insulin	42 (Ultra long)



Detemir	1-4	Peakless Insulin	12-24
Glargine	2-4	Peakless Insulin	20-24
NPH (Neutral protamine hagedorn)	2-4	4-10	10-16

SHORT-ACTING

Aspart (Rapid acting)	<0.25	0.5-1.5	3-5
Glulisine (Rapid acting)	<0.25	0.5-1.5	3-5
Lispro (Rapid acting)	<0.25	0.5-1.5	3-5
Regular -SC/IV	0.5-1.0	2-3	4-8 (DKA, Hyperglycemic hyperosmolar coma)
Inhaled human insulin	<0.5	1-2	3

INDICATIONS FOR INSULIN

00:56:36

- T2DM : HbA1c >10%, RPG > 250 mg/dL
- Catabolism
- Ketosis
- Pregnancy
- Organ damage (Metformin is not given if eGFR < 30 ml)
- Peri-operative
- HbA1C ≥ 9% despite 3 OHA
- LADA
- C-Peptide < 0.3

GLYCEMIC TARGETS

01:02:29

FPG	80-130 mg/dL
PPG	< 180 mg/dL
HbA1c	< 7% (<8% in elderly frail patients)

TIME IN RANGE

01:04:15

	RANGE	TIME
T1DM/T2DM	70-180 mg/dL	>70%
OLD/HIGH-RISK DM	70-180	>50%
GDM	70-180	>90%

T2DM MANAGEMENT

01:07:41

1. Medical nutrition therapy

- HbA1c: 1-3%
- Hypocaloric diet
- Carbohydrate restriction: low glycemic index- Oats
- Zero intake: Trans Fats

2. Exercise

- 30/min X 5 days - Brisk walking/Resistance training
- Increased GLUT 4: promotes sugar entry into the muscle
- Mild to moderate activity

	MECHANISM OF ACTION	EXAMPLES	HBA1C REDUCTION (%)	MAXIMUM DOSE	CONTRAINDICATIONS	SIDE EFFECTS
Biguanides	• Inhibits hepatic gluconeogenesis • ↑ peripheral utilization of sugar STOP METFORMIN WHEN • Radiographic contrast study • Hospitalized patient - NPO • Acidosis • Liver damage • Decompensated CHF	Metformin (Weight neutral drug)	1-2 %	2 grams/day	• eGFR < 30 ml/min → Triggers lactic acidosis	• Nausea • Vomiting • Metallic taste • Unexplained diarrhoea • Lactic acidosis • B12 deficiency
GLP-1 receptor agonists (Glucagon-like peptide 1 receptor agonist) - Parenteral/oral	• ↑ in insulin, ↓ glucagon levels • Delayed gastric emptying with early satiety → Weight loss • Stabilize atherosclerotic plaques - ↓ CV mortality	• Dulaglutide • Exenatide • Liraglutide • Lixisenatide • Semaglutide - once/weekly injection, oral tablet daily • These drugs have FDA black box warning - Risk of medullary carcinoma	• 0.5-1.0 • ↓ cardiovascular mortality			



		thyroid in rodents These are C/I in family history of MEN syndrome type II				
Dipeptidyl peptidase IV inhibitors	- Inhibit the degradation of native GLP-1 produced by ileum cells - Promotes Insulin secretion	- Linagliptin → Liver - Another drug metabolized by liver → Glipizide - Saxagliptin - Sitagliptin	0.5-0.8			
Insulin secretagogues: Sulphonylurea, Insulin ↑	Acts on ATP-sensitive potassium channels & depolarizes beta cells	- Glipizide - Glimeperide - Glyburide	0.5-0.8			Hypoglycemic attacks and weight gain
Mitiglinide	Insulin secretagogues but not sulfonylureas	- Nateglinide - Repaglinide	0.5-1.0			- Mild hypoglycemia - Diarrhoea - Vomiting
Thiazolidin-diones	↓ Insulin resistance by binding to PPAR gamma	- Pioglitazone - Rosiglitazone Advantage: - Reduce stroke recurrence	0.5-1.4		- Hepatic failure - CHF -CV mortality ↑ - Macular edema worsens - Induce ovulation	↑ Risk of bladder cancer



	- Promotes urinary loss of sugar - PCT - Concomitant loss of salt will reduce SBP by 3-6 mmHg - Cardioprotective - Don't use if eGFR < 30-45 ml/min - Reduce progression of CKD - Risk of UTI/Mycotic genital infection	- Empagliflozin - Canagliflozin - Dapagliflozin - Ertugliflozin				
α-Glucosidase inhibitors	Inhibits the GI absorption of sugar	- Acarbose - Voglibose	0.5-0.8			Osmotic diarrhoea

AGENTS USED FOR THE TREATMENT OF TYPE 1 OR TYPE 2 DIABETES

PARENTERAL

Amylin agonists	- Useful for T1 DM/T2DM - Regulates GI motility	Pramlintide	0.25-0.5
Insulin			Not limited
Medical nutrition therapy and physical activity		Low-calorie, carbohydrate-controlled diet, exercise	1-3

- Type 2 DM: HbA1c is between 8-10%

MANAGEMENT OF HYPERLIPIDEMIA IN DM

01:37:15

PRIMARY PREVENTION

	TARGET LDL
40-70 years - No cardiovascular risk factor - Moderate intensity ↑ cardiovascular risk factor - High intensity	< 100 mg/dL < 70 mg/dL



	TARGET LDL
• All ages with ASCVD on high-intensity statins	• < 55 mg/dL
LDL normal and Triglycerides > 500 mg/dL	Fenofibrate
LDL and Triglycerides elevated	Statin
LDL was elevated despite high-intensity statin, so the next drug was added	Ezetimibe followed by PCSK9 inhibitors

PCSK9 MONOCLONAL ANTIBODIES

- PCSK9 is an enzyme encoded on chromosome 1
- Plays a role in the regulation of the turnover of LDL receptors
- Inhibition will cause more LDL receptors available in hepatocytes to break down LDL
- Alirocumab
- Evolocumab
- Inclisiran

NON-ALCOHOLIC FATTY LIVER DISEASE

01:41:11

- Fatty infiltration >5% of hepatocytes affected
- Fatty infiltration + Inflammation/Hepatic ballooning without fibrosis: Non-alcoholic steatotic hepatitis
- The **Fib-4 score** has 4 parameters
 - Age
 - SGOT
 - SGPT
 - Platelet count
- Elastography or fibroscan - IOC
- Treatment: Diet control, weight loss (7-10%)

DIABETIC RETINOPATHY

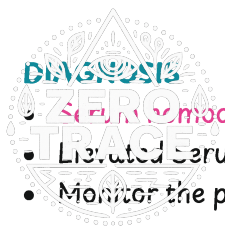
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- Earliest fundus finding - Microaneurysm in inner nuclear layer
- 25 times ↑ risk of blindness
- Non-proliferative Diabetic retinopathy - Macular edema (Visual loss)
- Proliferative Diabetic Retinopathy: Neovascularization → Blindness
 - Vitreous hemorrhage
 - Retinal detachment

TREATMENT

01:46:02

- Pan retinal photocoagulation
- Inj Bevacizumab (Anti-vascular endothelial growth factor)
- Type 1 DM → 5 years
- Type 2 DM → 15-20 years



01:47:12

- Serum cysteine levels - ↑
- Elevated Serum creatinine, eGFR
- Monitor the progression of retinopathy

DIABETIC NEPHROPATHY

01:48:28

- The earliest finding is **Hyperfiltration**
- M/c histological finding seen is **Diffuse glomerulosclerosis**
- M/c characteristic histological finding seen - **Kimmelstein Wilson change**

SCREENING TEST

01:49:20

- Urine albumin creatinine ratio: **30-300 mg/gram** → sign of early kidney damage.
- S.creatinine is normal
- Progression is the same as retinopathy
- For monitoring progression - eGFR, UAC

TREATMENT

01:50:55

- **ACE inhibitors + SGLT2 inhibitors and Finerenone**
- Stop Metformin if eGFR < 30 ml/min and start Insulin: 80% calculated dose

DIABETIC NEUROPATHY

01:52:44

- M/c long-term complication of Diabetes mellitus
- **Sensory** > Motor > Autonomic

SENSORY	MOTOR	AUTONOMIC
• Distal sensory loss • Unmyelinated C fibres are damaged • Pain/Paraesthesia in feet is perceived • Treatment - Duloxetine/Gabapentin	• 3rd nerve palsy with pupillary sparing	• Postural hypotension ◦ Supine to stand (wait for 3 mins in between) ◦ SBP < 20 mmHg ↓ ◦ Treatment - Midodrine

SKIN MANIFESTATIONS OF DM

01:56:11

DISEASES	DESCRIPTION
Diabetic dermopathy	• Brown macules on the shin of the patient 